

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Period: \_\_\_\_\_

**Keep in Mind**

**This is a practice test.** It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key. **100 points:** Part A 14, Part B 16, Part C 40, Part D 30.

**Part A — Vocabulary****14 questions, 1 point each**

**Set 1 — Samples and Bias.** Write the letter of the matching description next to each term.

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|----------------------------|-----------------------------------------------------------------------------------------|
| 1. _____ Population        | A. A number that describes a sample.                                                    |
| 2. _____ Parameter         | B. The list of individuals a sample can actually be drawn from.                         |
| 3. _____ Statistic         | C. Every individual the study wants to describe.                                        |
| 4. _____ Sampling frame    | D. A few whole groups are chosen at random, and everyone in them is studied.            |
| 5. _____ Stratified sample | E. Part of the population has no chance of being chosen at all.                         |
| 6. _____ Cluster sample    | F. A number that describes a population.                                                |
| 7. _____ Undercoverage     | G. The population is split into groups, and some are chosen at random from every group. |

**Set 2 — Studies and Experiments.** Same directions.

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|--------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 8. _____ Observational study   | A. The researcher assigns individuals to groups and imposes a condition.                                                 |
| 9. _____ Experiment            | B. A variable outside the study that could be moving both of the variables being compared.                               |
| 10. _____ Treatment            | C. The specific condition a group is given.                                                                              |
| 11. _____ Control group        | D. Individuals were chosen for the sample but never answered.                                                            |
| 12. _____ Lurking variable     | E. The researcher measures or surveys without assigning anyone to a group.                                               |
| 13. _____ Confounding variable | F. The groups inside the experiment differ in a second way too, so you cannot tell which difference produced the result. |
| 14. _____ Nonresponse          | G. The baseline group, which gets no treatment or the usual condition.                                                   |

**Part B — Multiple Choice****8 questions, 2 points each**

15. Riverbend High School records each student's **grade level** as 9, 10, 11, or 12. This variable is
- (A) quantitative, because the values are numbers.
  - (B) categorical, because the numbers are labels for groups.
  - (C) quantitative, because you can compute a mean of them.
  - (D) neither, because grade level is not really data.
16. District records show that **exactly 20% of all 1,500 Harbor Point members** swim before 9 a.m. That 20% is a
- (A) statistic, because it is written as a percent.
  - (B) statistic, because 1,500 is a large number.
  - (C) parameter, because it describes every member of the population.
  - (D) parameter, because it came out to a round number.

17. Which sentence states the difference between a stratified sample and a cluster sample?
  - (A) Stratified takes some from every group; cluster takes everyone from a few groups.
  - (B) Stratified takes everyone from a few groups; cluster takes some from every group.
  - (C) Stratified uses chance to choose; cluster does not use chance at all.
  - (D) There is no difference — the two names describe the same procedure.
18. A survey is conducted by calling landline telephones. Households that own no landline can never be reached. This is an example of
  - (A) nonresponse, because those households did not answer.
  - (B) undercoverage, because those households could never be chosen.
  - (C) sampling variability, because samples differ from each other.
  - (D) a response bias, because the question was worded badly.
19. A survey's method causes it to overestimate by 16 percentage points. The researcher repeats it using **ten times as many people** and the same method. The new estimate will most likely
  - (A) land very close to the true value, because the sample is large.
  - (B) underestimate, because large samples pull the other way.
  - (C) be impossible to predict in any direction.
  - (D) overestimate by about the same 16 points.
20. In a random sample of Riverbend students, those who use the Study Center pass *fewer* classes than those who do not. This result shows
  - (A) that the Study Center causes students to pass fewer classes.
  - (B) that passing classes causes students to avoid the Study Center.
  - (C) an association; a lurking variable could explain the whole thing.
  - (D) nothing at all, because the sample was chosen at random.
21. In an experiment, the purpose of the **control group** is to
  - (A) provide a baseline the treatment group can be compared against.
  - (B) give some subjects a break from the study.
  - (C) make the total number of subjects larger.
  - (D) remove the need for random assignment.
22. A researcher wants to know whether smoking causes lung disease in teenagers. No experiment is run. The best reason is that the experiment would be
  - (A) impractical, because volunteers are hard to find.
  - (B) unethical, because it would assign teenagers to smoke.
  - (C) impossible, because smoking cannot be measured.
  - (D) unnecessary, because observational studies prove causation.

### Part C — Short Answer & Computation

8 questions, 5 points each

23. **Bayside Middle School** has 750 students. For each student the principal records: grade level, minutes spent on homework last night, whether the student rides the bus, and favorite lunch entrée.
  - (a) Who are the **individuals** in this data set? \_\_\_\_\_
  - (b) Classify each variable.

| Variable                       | Categorical or quantitative? |
|--------------------------------|------------------------------|
| Grade level                    | _____                        |
| Minutes of homework last night | _____                        |
| Rides the bus (yes / no)       | _____                        |
| Favorite lunch entrée          | _____                        |

- (c) The principal's first draft question is "*How much homework?*" Explain why that is not a statistical question, then rewrite it so that it is.

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24. **Harbor Point Community Pool** has 1,500 members. The manager surveys 75 members chosen at random; 27 of them say they drive to the pool.

(a) Population: \_\_\_\_\_ Sample size: \_\_\_\_\_

Sample: \_\_\_\_\_

(b) What percent of the sample drives? About how many of all 1,500 members does that estimate?

(c) Is the 36% a parameter or a statistic? Say how you know.

\_\_\_\_\_

(d) Give one reason a **census** of all 1,500 members is impractical here.

\_\_\_\_\_

25. **Riverbend High School** has 900 students and wants a **stratified** random sample of 60, allocated in proportion to grade size.

(a) What fraction of the school is being sampled?

(b) Show the work for the **10th grade** row, then complete the table.

| Grade             | 9     | 10    | 11    | 12    | Total |
|-------------------|-------|-------|-------|-------|-------|
| Students in grade | 270   | 240   | 210   | 180   | 900   |
| Number sampled    | _____ | _____ | _____ | _____ | _____ |

26. **Cedar Ridge Apartments** has 600 households on a numbered list. The manager wants a **systematic** random sample of 60.

(a) Find the interval  $k$ .

(b) The random start is household 4. List the first four households selected:

\_\_\_\_\_

(c) Name the sampling technique each manager used.

| What the manager did                                                            | Technique |
|---------------------------------------------------------------------------------|-----------|
| Drew 6 of the 30 buildings at random, then surveyed every household in those 6. | _____     |
| Stood at the mailboxes on Tuesday and asked whoever walked past.                | _____     |
| Put all 600 household numbers in a generator and drew 60.                       | _____     |

27. **Millbrook Public Library** has 1,200 card holders and wants to know what fraction want Saturday evening hours. Two studies were run.

**Method 1 — clipboard at the front desk.** 90 people signed it; 72 said yes.

**Method 2 — random phone sample.** 120 card holders called; 54 said yes.

(a) Find the percent saying yes for each method.

(b) Estimate the number of all 1,200 card holders each method points to.

(c) One method is biased. Name it, name the **type** of bias, and say which **direction** it pushes the estimate.

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28. A random sample of 150 Riverbend students is taken. Of the 60 who use the **Study Center**, 39 passed all their classes. Of the 90 who do not use it, 72 passed all their classes. Nobody was assigned — students chose.

(a) Find each group's pass percent, and the gap between them.

(b) Explanatory variable: \_\_\_\_\_ Response: \_\_\_\_\_

(c) Using all 150 students, find the overall pass percent and estimate how many of all 900 students that is.

(d) Name one **lurking variable** that could produce this result, and say which way it would push the numbers.

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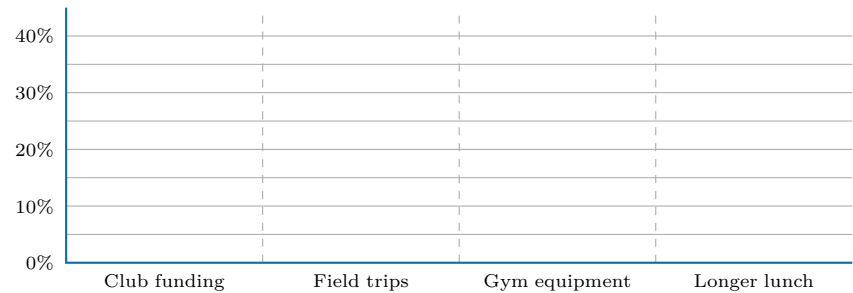
29. Riverbend now runs an **experiment**. Of 120 volunteers, a random number generator sends 60 to required weekly Study Center sessions and 60 to no sessions. Afterwards 45 of the session group and 33 of the other group passed all their classes.
- (a) Find each group's pass percent and the gap.

(b) Treatment: \_\_\_\_\_ Control group: \_\_\_\_\_

(c) Instead, Riverbend **blocks** on last quarter's grades: 80 volunteers passed everything and 40 did not. Randomly split each block in half. How many go to each group?

30. The Riverbend Student Council asked its stratified sample of 60 students how to spend a \$3,000 grant. Complete the table, then complete the bar graph.

| Choice        | Count | Relative frequency |
|---------------|-------|--------------------|
| Club funding  | 21    | _____              |
| Field trips   | 18    | _____              |
| Gym equipment | 15    | _____              |
| Longer lunch  | 6     | _____              |



About how many of all 900 Riverbend students would choose club funding?

Part D — Extended Response

3 questions, 10 points each

31. Riverbend has now run **two** studies of the same Study Center.
- Study A.** A random sample of students was watched. Of the students who *chose* to use the Study Center, 65% passed all classes; of those who did not, 80% did.
- Study B.** Of 120 volunteers, half were *assigned* to weekly sessions. Of those, 75% passed all classes; of the others, 55% did.

(a) Which study is observational and which is an experiment? Give the one feature that decides it.

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(b) Study A's gap is 15 points and Study B's is 20 points. Which study lets the council write "*the Study Center caused more students to pass*"? Justify your choice.

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(c) Explain how the same Study Center can look *harmful* in Study A and *helpful* in Study B.

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32. **Cedar Ridge Apartments** mailed a survey about a proposed dog park to all 600 households. 150 were returned, and 96 of those said yes. The management office's own sign-up records show that 240 of the 600 households actually want one.

(a) Find the response rate, the survey's percent saying yes, and the number of households that percent points to.

(b) The records say the truth is 240 of 600, or 40%. Name the bias, and give its size in percentage points *and* in households.

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(c) The manager mails again and gets 300 returns, 192 of them yes. Compute that percent and explain what the result shows.

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(d) Describe one change that would actually reduce the bias, and name the bias it removes.

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33. **Bayside Middle School** (750 students) wants to know whether to add a late activity bus. The office has a numbered list of all 750 students, and the four grades are believed to differ a lot in their answers.

(a) Choose **one** of the four sampling techniques and justify it from what this context allows.

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(b) Name one of the five **data collection methods** and say why it fits this question.

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(c) Instead, a teacher asks her own homeroom of 30 students; 12 say yes. She reports: " $12 \div 30 = 40\%$ , *and*  $0.40 \times 750 = 300$  students want the late bus." Her arithmetic is correct. Explain what is wrong with the report, and state what she *is* allowed to say.

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